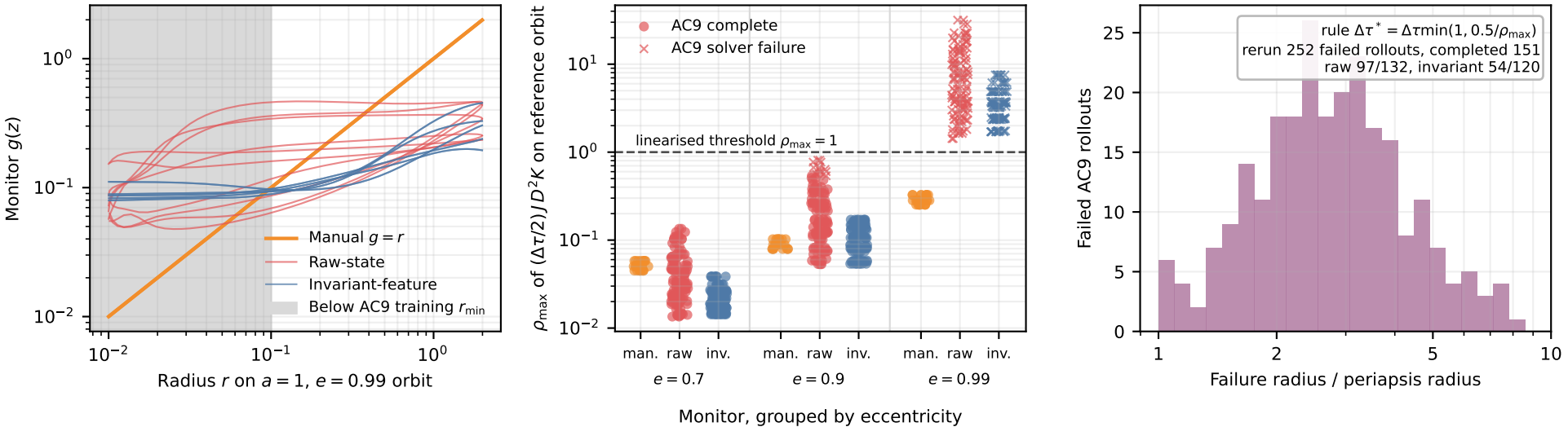

(a) Monitor extrapolation below training radius
Monitor $g(z)$
10^{0}
10^{-1}
10^{-2}
10^{-2}
10^{-1}
10^{0}
Radius r on $a = 1$, $e = 0.99$ orbit
Manual $g = r$
Raw-state
Invariant-feature
Below AC9 training $r_{\min}$
(b) Prospective contraction versus AC9 outcome
$\rho_{\max}$ of $(\Delta\tau/2)JD^2K$ on reference orbit
AC9 complete
AC9 solver failure
linearised threshold $\rho_{\max} = 1$
10^{1}
10^{0}
10^{-1}
10^{-2}
man. raw inv.
man. raw inv.
man. raw inv.
$e = 0.7$
$e = 0.9$
$e = 0.99$
Monitor, grouped by eccentricity
(c) Where failures occur; step-rule rerun
rule $\Delta\tau^{*} = \Delta\tau\min(1, 0.5/\rho_{\max})$
rerun 252 failed rollouts, completed 151
raw 97/132, invariant 54/120
Failed AC9 rollouts
0
5
10
15
20
25
1
2
5
10
Failure radius / periapsis radius